Complement Spectral-Mass Demand at the Minimal Bridge Clock

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Abstract

The exact odd deterministic anchors give a direct lower bound on the amount of modulus-complement spectral mass needed for relative coefficient matching. If all complement eigenvalues have modulus at most $q = 1/2$ and their squared mass is M, then relative matching at an odd order n forces M to grow like $(q_*/q)^n$. At the minimal tail-absorbed bridge slope $1/\log(10/7)$ this demand is $\sigma^{-0.9468615163\ldots}$, leaving only $0.0531384836\ldots$ of the available σ^{-1} mass exponent. This is a necessary condition under relative matching, not a proof that such matching occurs or fails.

1 Odd anchors and spectral cap

Let (μ_j) be a square-summable multiset with $|\mu_j| \le q = 1/2$, and put

$$M = \sum_j |\mu_j|^2, \qquad \tau_n = \sum_j \mu_j^n.$$

Then $|\tau_n| \le Mq^{n-2}$. For every odd $n \ge 3$, the deterministic anchor is exactly

$$a_n = \frac{q_*^n}{1+\lambda^{-n}}, \qquad q_* = (0.85\lambda)^{-1} = 0.700875225854775\ldots.$$

2 Mass-demand theorem

Theorem 2.1. *Let $n \ge 3$ be odd and $0 \le \theta < 1$. If*

$$|\tau_n - a_n| \le \theta a_n,$$

then

$$\boxed{M \ge \frac{(1-\theta)q^2}{1+\lambda^{-n}} \left(\frac{q_*}{q}\right)^n.}$$

Proof. The relative bound and positivity of a_n give $|\tau_n| \ge (1-\theta)a_n$. Combine this with $|\tau_n| \le Mq^{n-2}$ and rearrange. □

Corollary 2.2 (logarithmic clocks)**.** *Suppose $n_\sigma = a\log(1/\sigma) + O(1)$ is odd and the relative matching hypothesis of Theorem 2.1 holds with a fixed $\theta < 1$. Then*

$$\liminf_{\sigma\downarrow 0} \frac{\log M_\sigma}{\log(1/\sigma)} \ge a\log(q_*/q).$$

At

$$a_* = \frac{1}{\log(10/7)} = 2.803673252057129\ldots$$

the required exponent is

$$\gamma_* = a_* \log(q_*/q) = 0.9468615163684616\ldots.$$

The slope at which the demand reaches exponent one is

$$a_{\text{mass}} = \frac{1}{\log(q_*/q)} = 2.961017216974005\ldots.$$

Thus every slope strictly above a_{mass} is incompatible with the archived cap $M_\sigma \le \sigma^{-1}$ under uniform relative odd matching.

Proof. Take logarithms in Theorem 2.1; fixed factors and the $O(1)$ displacement of n_σ contribute $o(\log(1/\sigma))$. □

3 Protocol and exact scope

The computation evaluates the lower bound at odd orders $9, 19, 39$ with $\theta = 1/2$, and records q_*/q, γ_*, its slack $1 - \gamma_*$, and a_{mass}. No noisy coefficient is assumed or fitted.

Remark 3.1. Weighted-prefix convergence does not by itself imply relative matching at a moving odd order. The theorem therefore supplies a conditional necessary mass law, not a contradiction at the minimal clock. Gates A–E remain false/open; no operator identification or RH conclusion follows.